

Heralded Bell-Pair Generation in Satellite-Based Time-Bin and Hybrid Entanglement Distribution

Runzhe Mo^{*}, Huayue Gu[†], Ruozhou Yu[§], and Quntao Zhuang^{*†}

^{*}Ming Hsieh Department of Electrical and Computer Engineering, University of Southern California, Los Angeles, CA 90089, USA

[†]Department of Physics and Astronomy, University of Southern California, Los Angeles, California 90089, USA

[‡] Department of Information Technology, Kennesaw State University, Marietta, GA 30060, USA

[§]Department of Computer Science, North Carolina State University, Raleigh, NC 27695, USA

Email: runzhemo@usc.edu

Abstract—Satellite free-space links are a promising route to long-distance entanglement distribution, but different source placements and encodings may lead to distinct performance. We compare four representative architectures for producing heralded Bell pairs between remote ground stations through a mid-station satellite: low-brightness time-bin protocols and hybrid protocols based on two-mode squeezed vacuum (TMSV) states. We emphasize a feasibility distinction that is often hidden in rate comparisons: time-bin links require coherence-preserving nondestructive total-photon-number heralds or memory loading, while hybrid links use either quantum-scissor filtering in the satellite-source case or non-Gaussian entanglement swapping in the ground-source case. With these enabling operations specified, the four architectures can be compared through the trade-off between Bell-pair infidelity and generation probability.

Satellite-assisted links avoid the exponential distance penalty of terrestrial optical fiber by placing most of the optical propagation in free space. However, the satellite architecture is not unique. The entanglement source may be placed at ground stations, which simplifies source operation and maintenance, or on satellites, which can exploit downlink transmission to both users. In addition, the optical resource may be a time-bin Bell pair generated directly by the single-photon limit of spontaneous parametric down-conversion (SPDC) or a two-mode squeezed vacuum (TMSV) state used as a continuous-variable resource. We therefore compare four cases: time-bin ground-source, time-bin satellite-source, hybrid ground-source, and hybrid satellite-source. Here “hybrid” denotes a TMSV-based protocol whose accepted events are projected onto a single-rail qubit subspace.

We model both the uplink and downlink propagation as effective transmissivities η_{up} and η_{down} , with the downlink typically more favorable because atmospheric turbulence acts close to the receiver rather than close to the transmitter [1]. In our comparison, we take uplink loss to be 20 dB and downlink loss to be 10 dB, where an L dB loss corresponds to $\eta = 10^{-L/10}$, as a proof-of-principle example [2].

Time-Bin Model and Heralded Filtering. For the time-bin architectures, we use a time-bin encoded SPDC source as the entanglement generation source. Each photon occupies one of two temporal modes, early e or late l , and the qubit is encoded in the coherent superposition of these two time bins. In the

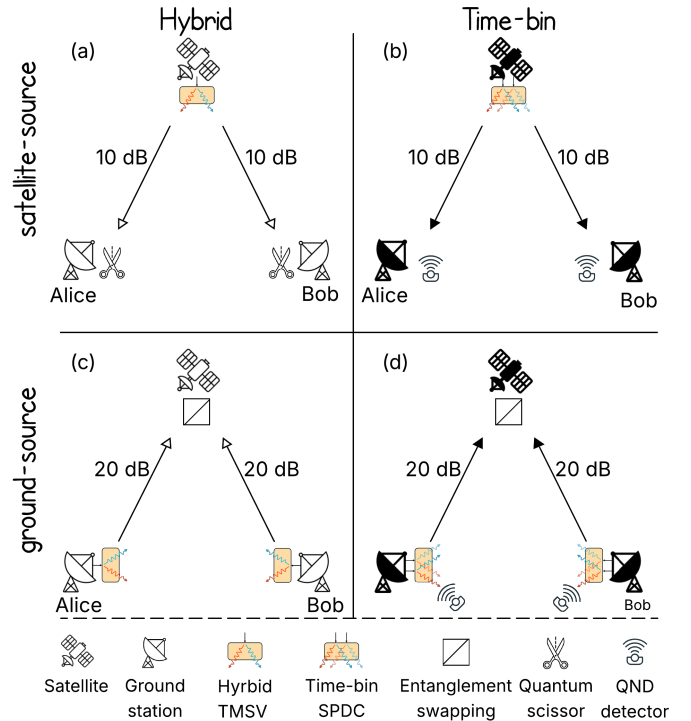


Fig. 1. Four architectures: (a) hybrid satellite source with quantum scissors (QS); (b) time-bin satellite source with QND arrival detection; (c) hybrid ground source with threshold BSM; and (d) time-bin ground source with BSM and QND.

low-brightness regime,

$$|\psi_{\text{tb}}\rangle \simeq |\text{vac}\rangle + \lambda S^\dagger |\text{vac}\rangle + \frac{\lambda^2}{2} (S^\dagger)^2 |\text{vac}\rangle + O(\lambda^3). \quad (1)$$

Here $S^\dagger = a_e^\dagger b_e^\dagger + a_l^\dagger b_l^\dagger$ and λ is the squeezing parameter. The first-order term prepares the desired single-pair time-bin Bell state, while the second-order term contains two-pair components in the same or different time bins. They are the dominant source of multiphoton contamination in the later swapping analysis. Increasing λ raises the raw probability of generating non-vacuum emission events, but it also increases the relative weight of multipair components.

Ground-source schemes transmit two independently generated modes through uplinks for swapping, whereas satellite-source schemes distribute one onboard pair directly through two downlinks (shown in Fig. 1).

The multiphoton sector in the time-bin source severely limits raw Bell-pair generation. Without heralded storage or filtering on the retained modes, accepted events can include false heralds where multipair emissions, together with vacuum components in the output arms, mimic the desired single-pair Bell event. In the unfiltered model considered here, this ambiguity upper bounds the raw Bell-pair fidelity at 50% [3]. This bound applies only to raw Bell-pair generation for future use; for immediate measurements, such as verification or quantum key distribution (QKD), vacuum events can be removed during postselection or sifting. Therefore, high-fidelity heralded Bell-pair generation requires a nondestructive total-photon-number measurement, e.g., a quantum non-demolition (QND) arrival measurement or heralded memory loading, that projects each retained time-bin qubit onto the single-photon sector $n_e + n_l = 1$ without distinguishing the early and late modes. Independent nondestructive measurements of the early and late modes would reveal which-time-bin information and therefore are not sufficient for preserving Bell-pair coherence.

Hybrid Model and Heralded Conversion. We model the hybrid entanglement source as a single-temporal-mode TMSV state. Satellite-source TMSV states are projected onto single-rail qubits using ideal heralded quantum scissors after transmission [4]; their detector-level implementation is not modeled. For ground sources, Gaussian swapping [5], [6] is ineffective in the relevant loss regime: the Gaussian Peres–Horodecki criterion requires symmetric measured-arm transmissivity $\eta > 1/2$ for the swapped state to be entangled [7]. We therefore employ non-Gaussian DV entanglement swapping at the middle station, which both swaps the entanglement and conditionally filters the retained modes into the single-rail qubit subspace, with the accepted detection patterns determining the success probability and Bell-state fidelity.

Main Result. Since all four architectures rely on a probabilistic heralding event, we compare them by the success probability of producing an accepted Bell pair and the infidelity $1 - F$ of the retained two-qubit state. For Fig. 2, we propagate the sources through pure-loss channels in the Gaussian description, truncate to a finite Fock basis before the non-Gaussian measurements, and then apply the time-bin swapping/filtering or ideal quantum-scissor maps. The curves sweep the source squeezing; scissor-assisted points are additionally optimized over the scissor gain.

We use on-off detection throughout; ideal PNR filtering would remove the time-bin multiphoton ambiguity and yield vanishing infidelity in our model.

In the time-bin ground-source architecture, the rate is suppressed because two probabilistic SPDC sources must emit usable pairs simultaneously and the transmitted arms propagate through lossy uplinks. The time-bin satellite-source architecture avoids both penalties by using a single onboard source and downlink distribution. In both time-bin cases,

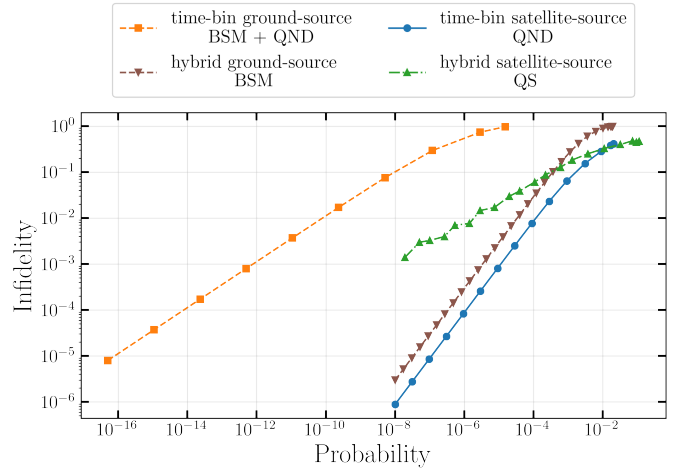


Fig. 2. Bell-pair infidelity $1 - F$ versus success probability at $\eta_{\text{down}} = 0.1$ and $\eta_{\text{up}} = 0.01$. Markers vary source squeezing r ; the hybrid satellite-source gain g is optimized at each r .

increasing λ raises the accepted-event probability but also increases multiphoton contributions under on-off detection.

The hybrid architectures show a different tradeoff. The hybrid satellite-source protocol benefits from downlinks, but both lossy arms must be converted by probabilistic quantum scissors, which compresses the heralding probability and can degrade the retained fidelity. The hybrid ground-source protocol instead uses deterministic TMSV generation and threshold-detection swapping; only the measured arms experience the uplink, so loss mainly changes the swapping quality rather than directly adding noise to the retained arms. At a comparable rate, the hybrid ground-source protocol reaches a Bell-pair infidelity within the same order of magnitude as the time-bin satellite-source protocol, while avoiding both coherence-preserving total-photon-number QND/memory loading and the quantum-scissor filtering.

Discussion. From a feasibility perspective, time-bin schemes require coherence-preserving QND detection or heralded memory loading, whereas the hybrid satellite-source scheme requires quantum scissors after both downlinks. The hybrid ground-source scheme instead combines swapping and single-rail filtering in one middle-station DV measurement, avoiding both operations and thus offering the most direct implementation under our idealized assumptions. Detector inefficiency, dark counts, mode mismatch, and phase instability remain to be assessed.

A promising future direction is bosonic-code-assisted CV–DV entanglement distribution, where a discrete logical Bell pair is embedded in a loss-aware bosonic encoding [8], [9]. Such encoding may convert photon loss into heralded erasures or correctable logical errors, although it cannot evade the fundamental repeaterless rate–loss bound [10]. Future work should assess whether this protection improves satellite entanglement rates after accounting for encoding and recovery overhead.

ACKNOWLEDGMENTS

Mo and Zhuang were supported in part by NSF grant No. 2350153. Gu and Yu were supported in part by NSF grant No. 2350152. The information reported herein does not reflect the position or the policy of the funding agencies.

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